

HOMEWORK 1: EXPLORATORY ANALYSIS OF A MUSIC CATALOG

CS 460200 INTRODUCTION TO MACHINE LEARNING (FALL 2026)

<https://george930502.github.io/IntroML-f26/>OUT: **Thursday, September 24** PRE-REGISTRATION DUE: **Monday, September 28**REPORT DUE: **Thursday, October 8** DEMO: **week of November 10**

POINTS: report 50, demo 50

Summary You are given a catalog of 114,000 music tracks. Each track is described by audio features that an automatic analyzer produced from the recording, together with a genre label. The team you are working with wants to build a genre tagger and a hit predictor on top of this catalog. Your task is to establish what the catalog can support before anyone trains such a model on it. You will audit the file for defects, describe the geometry of the feature space with PCA, test whether the catalog contains real structure, fit regression models to two targets of very different difficulty, and state what the catalog does and does not support. The later assignment builds on the ground prepared here, but everything you hand in for this one concerns this catalog and nothing else. Points are awarded for evidence and justification, not for model accuracy. Several questions below have little or no signal in this data; establishing that, and explaining why, earns full credit.

Deadlines and deliverables

Monday, September 28	Pre-registration, at most one page (Section 2).
Thursday, October 8	Report (PDF, at most 6 pages and 8 figures), notebook, and AI-use log (Section 8).
Week of November 10	Demo (Section 9).

Start Here: Instructions

- **Collaboration policy.** Work individually. You may discuss concepts with classmates, but the analysis, the code, and the writing you submit must be your own.
- **Use of AI tools.** Permitted, including coding agents. You must submit the AI-use log described in Section 8, and you must be able to explain and modify any part of your submission at the demo.
- **Submitting your work.** Submit on eclass. The pre-registration in Section 2 is a separate, earlier submission. The main submission consists of three items: the report as a PDF of at most six pages and at most eight figures; the notebook, which must run from top to bottom; and the AI-use log, which does not count toward the page limit. Give your report the same section titles as this handout, in the same order.
- **Late submission.** See the course policy on eclass. The pre-registration is the one exception: it earns no credit after its deadline, for the reason given in Section 2.
- **Materials.** The data file `tracks.csv` and a starter notebook are posted on eclass. The starter notebook loads the file and works through one audit finding as an example of the level of evidence expected in Section 3.
- **Environment.** Google Colab is sufficient; no GPU is needed. You may use `pandas`, `numpy`, `scipy`, `matplotlib`, and `scikit-learn`.

- **Scope.** Nothing in this assignment requires material taught after October 8. Cross-validation, ROC analysis, and the classifiers introduced later in the course are neither required nor rewarded here.

1 The Data

The file `tracks.csv` contains one row per track and genre listing, with the following columns.

Column	Type	Description
<code>track_id</code> , <code>track_name</code> , <code>artists</code> , <code>album_name</code>	text	identifiers and metadata
<code>track_genre</code>	text	the playlist category from which the row was collected
<code>popularity</code>	0–100	a score computed by the streaming platform
<code>duration_ms</code>	integer	track length in milliseconds
<code>explicit</code>	boolean	explicit lyrics flag
<code>danceability</code> , <code>energy</code> , <code>speechiness</code> , <code>acousticness</code> , <code>instrumentalness</code> , <code>liveness</code> , <code>valence</code>	0–1	audio descriptors produced by an automatic analyzer
<code>loudness</code>	dB	typically negative
<code>tempo</code>	BPM	estimated beats per minute
<code>key</code> , <code>mode</code> , <code>time_signature</code>	integer	musical attributes stored as integer codes

The provider states neither that every value in the file was measured successfully, nor that every row describes a different track. Do not assume either.

2 Pre-Registration (4 points)

Submit this section by **Monday, September 28**, before you begin the analysis. Read the column descriptions in Section 1 and inspect no more than the first 50 rows of the file. In at most one page, state the following.

1. (2 points) Three claims you expect to hold in this catalog. For each claim, state the observation that would show it to be false.
2. (1 point) Your predicted held-out root-mean-square error for the two regression targets of Section 6, in the units of each target, with one sentence of reasoning for each.
3. (1 point) Two defects you expect to find in the file, and one question you would want answered before trusting a model trained on it.

Points are awarded for completeness and specificity, not for being right. You will compare these statements with your results in Sections 6 and 7, and you will be asked about them at the demo. **A pre-registration submitted after the deadline earns no credit**, because its purpose is to record your expectations before you know the answers.

3 Auditing the File (12 points)

Automatically collected catalogs contain defects: values that were never measured, rows that appear more than once, and fields that mean something other than what their name suggests. Find the defects in this file that would mislead someone who trains a model on it.

1. (8 points) Report at most eight findings, ordered by how much each would distort a model trained on the file. For each finding, give the evidence, in the form of one number or one figure; the mechanism you believe produced it; the action you take; and what that action costs you.
2. (2 points) Explain why your first finding outranks your second.
3. (2 points) Carry out the decisions you took, as code in your notebook that turns the raw table into the table the rest of your analysis uses. Findings you decided not to act on belong in the report as stated limitations of your analysis, rather than in the code.

An unordered list of every unusual value in the file, with no mechanism and no ranking, earns partial credit at best.

4 The Geometry of the Feature Space (10 points)

The numeric audio columns define a feature space. Describe its shape.

1. (3 points) How many dimensions does this space effectively have? State the criterion you applied and the evidence for it.
2. (3 points) Interpret the leading principal components. Name each one in plain language and give the loadings that support the name.
3. (3 points) Report what happens when the features are not standardized before PCA. Account for the result in terms of the quantity that PCA maximizes, rather than in terms of intuition: a short derivation, or a variance calculation on your own numbers, is expected here.
4. (1 point) Identify the columns in Section 1 that should not enter this analysis as numeric variables, and state what you did with them.

5 Structure in the Catalog (10 points)

The team believes that the catalog falls into natural kinds of music. Test that belief at two levels of aggregation.

1. (4 points) **Genres.** Represent each genre by the profile of its tracks and group the genres by hierarchical clustering. Compare two dissimilarity measures, or two linkage rules. Report where the two groupings disagree, and state which of them you would defend.
2. (3 points) **Tracks.** Cluster individual tracks. Justify the number of clusters with at least two criteria, and describe in plain language what each cluster contains. Not every method scales to 114,000 rows; if you cluster a sample, or evaluate on one, say which sample and why.
3. (3 points) Answer the following objection with evidence: *these clusters are not musical structure; they are an artifact of the way the file was assembled.* A comparison against the genre labels is one form of evidence, and not the only one.

6 Two Regression Targets (10 points)

The team wants to know which of the following questions the audio features can answer: how energetic a track is, and how popular a track will be. Fit a linear model for each.

1. (2 points) Restate your predictions from Section 2, then report the held-out root-mean-square error for both targets, together with the error of a model that ignores the features and always predicts the training mean.

2. (3 points) Account for the difference between the two targets. Explain what makes the harder one hard, using evidence from this data rather than a general statement about noise.
3. (3 points) If the audio features do not explain the harder target, look for a variable in the file that does. Identify a candidate and support it with a measurement.
4. (2 points) Fit the model again with regularization, on a set of features that you have reason to believe are related to one another. Vary the penalty over several orders of magnitude and report what happens to the coefficients and to the held-out error. If the penalty does not improve the error, say so and explain why not, then find a setting in which it does: a smaller training sample, a richer set of basis functions, or both.

Close this section with one sentence of the following form: the audio features support X, and do not support Y.

7 Conclusions (4 points)

Close the report with a summary that a reader could act on without reading the rest of it. At most one page.

1. (1 point) State what your cleaning keeps and what it discards.
2. (1 point) Give three claims about this catalog that your analysis supports, each with the evidence behind it and the observation that would refute it.
3. (1 point) Name the property of this file that would make an evaluation based on a random split of rows look better than the model deserves, and give the evidence for that property from your audit.
4. (1 point) Fit a logistic regression for a two-class question of your choice, using a training and test split. Choose a question whose two classes are of comparable size, so that the comparison below carries information. Report the fraction of test cases it classifies correctly, next to the fraction obtained by always predicting the more common class, and state what the comparison tells you about the catalog.

State which of your claims from Section 2 did not survive the analysis, and what changed your mind.

8 AI-Use Log

Submit at most one page recording which parts of the work you delegated to an AI tool, how you checked its output, and at least one case in which it produced something wrong or misleading, including how you noticed. The log carries no points of its own. It is the basis of one of the demo questions, so it must describe what you actually did.

9 Demo (50 points)

The demo takes place in the week of **November 10**, in a session that also covers the next assignment. The schedule and the length of each slot are announced on eclass. Bring your report and a notebook you can run. Questions are drawn from a bank and refer to your own submission. Typical questions are:

- Take one decision from your audit. What changes further down the analysis if you decide the other way? Show the effect in your notebook.
- Your report interprets the first principal component. Suppose one column were multiplied by 1000. What happens to that interpretation, and why?

- You reported two regression results. Which of them would you rely on, and what evidence would change your mind?
- Reproduce the error recorded in your AI-use log.
- A TA changes one element of your notebook, such as a random seed, a scaling step, or a filter. State what you expect to happen and why, then run it.

Points are awarded for explaining, defending, and modifying your own work, and not for the work being free of mistakes.

10 Grading

Report (50 points)

Section	Points
2 · Pre-registration	4
3 · Auditing the file	12
4 · Geometry of the feature space	10
5 · Structure in the catalog	10
6 · Two regression targets	10
7 · Conclusions	4

Demo (50 points)

Criterion	Points
Explains and defends own decisions	15
Predicts the effect of a change, with reasons	15
Argues from definitions when questioned	10
Accounts for the AI tools used	10

Open-ended answers are marked in four bands.

Band	Description	Credit
1	A result or a number is reported.	25%
2	Several results are reported without a connection between them.	50%
3	Evidence, mechanism, and decision are connected into one argument.	85%
4	The argument is also tested against an alternative explanation, or its limits are stated.	100%

Correct observations that are neither ordered nor explained fall in band 2, regardless of how many a report contains.

11 Reading

The lecture slides are the primary material for every section of this assignment. The textbook sections below cover the same topics in more depth and are optional; read the named sections only.

Section	Lecture slides	Kubat, <i>An Introduction to Machine Learning</i>	Bishop, <i>Pattern Recognition and Machine Learning</i>
3 · Audit	L1, L12	§1.3 Problems with Available Data, p. 6: attribute types, irrelevant and redundant attributes, missing values, inconsistent data, and noise. §11.5 Unknown Attribute Values, p. 222.	—
4 · Geometry	L12	§11.6 Attribute Selection, p. 224, on selection against extraction. Kubat does not cover PCA.	§12.1.1 and §12.1.2, p. 561–565, the two formulations of PCA used in the slides. §1.4 The Curse of Dimensionality, p. 33.
5 · Structure	L11	§15.1 Cluster Analysis, p. 297. §15.2 k-Means, p. 301, including the normalization of attributes. §15.3, p. 305. §15.4 Hierarchical Aggregation, p. 307.	§9.1 K-means Clustering, p. 424.
6 · Regression	L1, L2	—	§1.1, p. 4–11, where held-out error is measured by E_{RMS} , equation (1.3). §3.1.4 Regularized Least Squares, p. 144.
7 · Conclusions	L2	—	§4.3.2 Logistic Regression, p. 205.

12 Data Source

Spotify Tracks Dataset, distributed on Hugging Face as [maharshipandya/spotify-tracks-dataset](https://huggingface.co/maharshipandya/spotify-tracks-dataset), used here for coursework. Cite this source in your report.